

Condensed Matter Physics

Assignment - 3

1. Consider a one-dimensional periodic potential consisting of a sequence of rectangular barriers of height U , width b , and period a :

$$V(x) = \begin{cases} U, & 0 \leq x < b \pmod{a}, \\ 0, & b \leq x < a \pmod{a}. \end{cases} \quad (1)$$

- (a) Starting from the time-independent Schrödinger equation and applying Bloch's theorem, derive the dispersion relation for the Kronig-Penney potential.
- (b) Choosing values for U , b , a so that ($0 < b < a$), numerically solve the Kronig-Penney dispersion to obtain $E(\mathbf{k})$ within the first Brillouin zone. Plot the dispersion and clearly indicate the allowed bands and band gaps. Mark the lowest two energy bands and the zone boundaries at $k = \pm\pi/a$.
- (c) Consider the limiting case of the Kronig-Penney model where $b \rightarrow 0$ and $U \rightarrow \infty$, while keeping the product $V_0 \equiv Ub$ constant. In this limit, the potential reduces to a Dirac comb, and the dispersion relation takes the form

$$\cos(ka) = \cos(qa) + \frac{maV_0}{\hbar^2} \frac{\sin(qa)}{qa}, \quad (2)$$

where q is the wave vector inside the potential wells.

Now, by allowing the wave vector k to take complex values, numerically demonstrate that the corresponding energy solutions lie within the forbidden energy gaps of the band structure. (It is sufficient to perform the numerical analysis within the first energy band gap.)

2. Consider a two-dimensional ~~square lattice with lattice~~ constant a , where each site hosts a single orbital $|m, n\rangle$, with m and n denoting the lattice indices along the x and y -directions, respectively. Electrons can hop between nearest neighbours with hopping amplitude t , and each site has an on-site energy ϵ_0 . The tight-binding Hamiltonian is given by:

$$H = -t \sum_{m,n} \left(|m+1, n\rangle \langle m, n| + |m, n+1\rangle \langle m, n| + \text{h.c.} \right) + \epsilon_0 \sum_{m,n} |m, n\rangle \langle m, n|. \quad (3)$$

- (a) Derive the dispersion relation $E(\mathbf{k})$ using Bloch's theorem, and plot the equal-energy contours in the $k_x - k_y$ plane within the first Brillouin zone, i.e., for $-\pi/a < k_x < \pi/a$ and $-\pi/a < k_y < \pi/a$.
- (b) Compute the density of states (DOS) corresponding to the energy band. Examine if there are any singularities that appear, and if so, comment on the nature of these van Hove singularities.
- (c) Consider each site is populated on average by a single electron (half-filled case) and determine the Fermi energy E_F and the corresponding Fermi momentum $\mathbf{k}_F$. Plot the Fermi surface in the first Brillouin zone.

3. Consider the one-dimensional tight-binding Hamiltonian that includes an additional modulated on-site potential term,

$$\hat{H} = -t \sum_n (|n\rangle \langle n+1| + \text{h.c.}) + U \sum_n |n\rangle \langle n| + V \sum_n (-1)^n |n\rangle \langle n|. \quad (4)$$

- (a) Determine the dispersion relation corresponding to this Hamiltonian.
- (b) Further, demonstrate that the Brillouin zone becomes reduced as a result of the staggered potential.

Hint:- You may divide the lattice into two sublattices (denoted A and B) and perform the calculation accordingly, or equivalently, consider two momentum modes C_k and $C_{k+\pi}$.

4. Consider a one-dimensional lattice composed of two parallel chains, where each chain corresponds to a distinct orbital — the lower chain represents the p -orbital and the upper chain represents the s -orbital. Electrons can hop between neighbouring sites within the same chain, as well as between the two orbitals on the same site. Such a system can be described by the following tight-binding Hamiltonian:

$$H = -t \sum_n (|n, s\rangle \langle n+1, s| + \text{h.c.}) - t' \sum_n (|n, p\rangle \langle n+1, p| + \text{h.c.}) - \Omega \sum_n (|n, s\rangle \langle n, p| + \text{h.c.}) + \epsilon_s \sum_n |n, s\rangle \langle n, s| + \epsilon_p \sum_n |n, p\rangle \langle n, p|. \quad (5)$$

Here, t and t' denote the hopping amplitudes along the s and p orbital chains, respectively, Ω represents the inter-orbital coupling between the two chains, and ϵ_s and ϵ_p are the on-site energies of the respective orbitals.

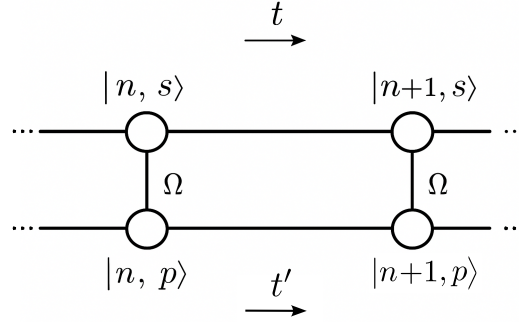


Figure 1: Schematic representation of two coupled 1D chains corresponding to s and p orbitals. The hopping amplitudes along the s and p chains are t and t' respectively, while the on-site inter-chain coupling (hybridization) is Ω .

- (a) Perform a Fourier transform to momentum space using the Bloch states

$$|k, \sigma\rangle = \frac{1}{\sqrt{N}} \sum_n e^{ikna} |n, \sigma\rangle, \quad (6)$$

where $\sigma = s, p$, and express the Hamiltonian as a 2×2 matrix $H(k)$.

- (b) Obtain the energy dispersion relation $E_{\pm}(k)$ and sketch the two resulting energy bands for representative parameters ($t' < t$, $\Omega \neq 0$).

5. The wavefunction of a particle in a periodic lattice with lattice constant 'a' obeys Bloch's theorem,

$$\psi_{n,k}(x) = e^{ikx} u_{n,k}(x),$$

where $u_{n,k}(x)$ has the periodicity of the lattice.

- (a) Substituting the Bloch form into the time-independent Schrödinger equation, show that the problem reduces to

$$h_k u_{n,k}(x) = \epsilon_{n,k} u_{n,k}(x), \quad \text{where} \quad h_k = \frac{1}{2m} (p_x + \hbar k)^2 + V(x),$$

where the periodic potential can be expanded as

$$V(x) = \sum_G V_G e^{iGx},$$

where $Ga = 2\pi n$ are the reciprocal lattice vectors.

- (b) Treating the potential $V(x)$ as a weak perturbation, only the lowest harmonic is important

$$V(x) \approx V_{G_1} e^{iG_1 x} + V_{G_1}^* e^{-iG_1 x}.$$

(Since the potential is real, $V_G^* = V_{-G}$). Show that a gap of width $\Delta = 2|V_{G_1}|$ opens at the Brillouin zone boundary $k = G_1/2$ (i.e., $k = \pi/a$ for the first zone). Estimate Δ from the truncated potential.

- (c) For an optical lattice with a periodic Gaussian potential

$$V(x) = -V_0 \sum_n e^{-\alpha(x-na)^2},$$

compute the Fourier coefficients V_G and hence estimate the band gap at the first zone boundary in the weak-potential limit.

- (d) Extend the Bloch wave-vector into the complex plane as

$$k = \frac{G}{2} + i\chi, \quad \chi \in \mathcal{R},$$

and repeat (b) to show that the allowed energies can exist within the forbidden region Δ , by suitably choosing χ .

6. Consider a body-centered cubic (BCC) lattice with lattice constant a . Identify the primitive lattice vectors, determine the corresponding reciprocal lattice vectors, and identify the type of reciprocal lattice obtained. Repeat the calculation for a two-dimensional equilateral triangular lattice with lattice constant a .

7. Ultracold atoms confined in an optical lattice experience a periodic potential given by

$$V(x) = V_0 \cos^2\left(\frac{\pi x}{a}\right), \quad (7)$$

where V_0 is the lattice depth and a is the lattice spacing.

- (a) Construct the Bloch wave function $\Psi(x)$ for a particle in this lattice within the tight-binding approximation.
- (b) Estimate the nearest-neighbour tunnelling amplitude t , which characterises the hopping of an atom between adjacent lattice sites.